

ident
integer_literal
infix_symbol
lowercase_ident
capitalized_ident

<i>infix_op</i>	$::=$ =	
<i>typeconstr_name</i>	$::=$ <i>lowercase_ident</i>	
<i>typeconstr</i>	$::=$ unit bool int	
<i>value_name</i> , <i>x</i>	$::=$ <i>lowercase_ident</i>	
<i>typeexpr</i> , <i>t</i>	$::=$ <i>'ident</i> (<i>typeexpr</i>) <i>typeexpr</i> $\rightarrow$ <i>typeexpr'</i> <i>typeconstr</i>	S
<i>constant</i>	$::=$ <i>integer_literal</i> false true ()	
<i>expr</i> , <i>e</i>	$::=$ <i>value_name</i> <i>constant</i> (<i>expr</i>) <i>expr</i> <i>expr'</i> <i>expr</i> <i>infix_op</i> <i>expr'</i> if <i>expr</i> then <i>expr</i> ₁ else <i>expr</i> ₂ <i>expr</i> ; <i>expr'</i> function <i>value_name</i> $\rightarrow$ <i>expr</i> let <i>value_name</i> = <i>expr</i> in <i>expr'</i> <i>C.e</i> { <i>v/x</i> } <i>e</i>	S S M M
<i>Value</i> , <i>v</i>	$::=$ <i>constant</i> function <i>value_name</i> $\rightarrow$ <i>expr</i>	
<i>C_expr</i> , <i>C</i>	$::=$ <i>_expr</i> <i>v _</i> <i>_infix_op</i> <i>expr</i> <i>v</i> <i>infix_op</i> <i>_</i> if <i>_</i> then <i>expr</i> else <i>expr'</i>	

		$_ ; \text{expr}$
		let $\text{value_name} = _ \text{in expr}$
E	$::=$	
		empty
		$E, \text{value_name} : \text{typeexpr}$
		$E, \text{value_name} : \text{typeexpr}, E' \quad \mathbf{M}$
formula	$::=$	
		judgement
		not (formula)
		$E = E'$
terminals	$::=$	
		$\star$
		$\rightarrow$
		function
		$-$
		$\vdash$
		$\longrightarrow$
		$\{$
		$\}$
$J\text{type}$	$::=$	
		$\text{value_name} \text{ in dom } (E)$
		$E \vdash \mathbf{ok}$
		$\vdash \text{infix_op} : t \rightarrow t' \rightarrow t''$
		$\vdash \text{typeconstr} : \mathbf{Type}$
		$E \vdash \text{typeexpr} : \mathbf{Type}$
		$E \vdash \text{constant} : t$
		$E \vdash e : t$
$J\text{op}$	$::=$	
		$e \longrightarrow e'$
judgement	$::=$	
		$J\text{type}$
		$J\text{op}$
user_syntax	$::=$	
		ident
		integer_literal
		infix_symbol
		lowercase_ident
		capitalized_ident
		infix_op
		typeconstr_name
		typeconstr
		value_name

$typeexpr$
 $constant$
 $expr$
 $Value$
 C_expr
 E
 $formula$
 $terminals$

$value_name \text{ in dom } (E)$

$\frac{}{value_name \text{ in dom } (E, value_name : typeexpr)} \quad \text{VID_VN1}$

$\frac{value_name \text{ in dom } (E)}{value_name \text{ in dom } (E, value_name' : typeexpr)} \quad \text{VID_VN2}$

$E \vdash \text{ok}$

$\frac{E \vdash typeexpr : \mathbf{Type} \quad \mathbf{not} (value_name \text{ in dom } (E))}{E, value_name : typeexpr \vdash \text{ok}} \quad \text{EOK_VALUE_NAME}$

$\vdash infix_op : t \rightarrow t' \rightarrow t''$

$\frac{}{\vdash=: \mathbf{bool} \rightarrow \mathbf{bool} \rightarrow \mathbf{bool}} \quad \text{INFIX_OP_EQUAL}$

$\vdash typeconstr : \mathbf{Type}$

$\frac{}{\vdash \mathbf{unit} : \mathbf{Type}} \quad \text{TYPCONSTR_UNIT}$

$\frac{}{\vdash \mathbf{bool} : \mathbf{Type}} \quad \text{TYPCONSTR_BOOL}$

$\frac{}{\vdash \mathbf{int} : \mathbf{Type}} \quad \text{TYPCONSTR_INT}$

$E \vdash typeexpr : \mathbf{Type}$

$\frac{E \vdash \text{ok}}{E \vdash' ident : \mathbf{Type}} \quad \text{TYPEEXPR_IDENT}$

$\frac{E \vdash typeexpr : \mathbf{Type}}{E \vdash (typeexpr) : \mathbf{Type}} \quad \text{TYPEEXPR_PAREN}$

$\frac{E \vdash typeexpr : \mathbf{Type} \quad E \vdash typeexpr' : \mathbf{Type}}{E \vdash typeexpr \rightarrow typeexpr' : \mathbf{Type}} \quad \text{TYPEEXPR_ARROW}$

$\frac{\vdash typeconstr : \mathbf{Type}}{E \vdash typeconstr : \mathbf{Type}} \quad \text{TYPEEXPR_CONSTR0}$

$E \vdash constant : t$

$\frac{E \vdash \text{ok}}{E \vdash integer_literal : \mathbf{int}} \quad \text{CONSTANT_INT}$

$$\begin{array}{c}
\frac{E \vdash \mathbf{ok}}{E \vdash \mathbf{false} : \mathbf{bool}} \quad \text{CONSTANT_FALSE} \\
\frac{E \vdash \mathbf{ok}}{E \vdash \mathbf{true} : \mathbf{bool}} \quad \text{CONSTANT_TRUE} \\
\frac{E \vdash \mathbf{ok}}{E \vdash () : \mathbf{unit}} \quad \text{CONSTANT_UNIT}
\end{array}$$

$$\boxed{E \vdash e : t}$$

$$\begin{array}{c}
\frac{E \vdash \mathbf{ok} \quad \mathbf{not}(x \text{ in dom}(E_2)) \quad E = E_1, x : t, E_2}{E \vdash x : t} \quad \text{EET_VALUE_NAME} \\
\frac{E \vdash \mathbf{ok} \quad E \vdash \text{constant} : t}{E \vdash \text{constant} : t} \quad \text{EET_CONSTANT} \\
\frac{E \vdash e : t}{E \vdash (e) : t} \quad \text{EET_PAREN} \\
\frac{E \vdash e : t_1 \rightarrow t_2 \quad E \vdash e' : t_1}{E \vdash e \ e' : t_2} \quad \text{EET_APPLY} \\
\frac{\vdash \text{infix_op} : t \rightarrow t' \rightarrow t'' \quad E \vdash e : t \quad E \vdash e' : t'}{E \vdash e \text{ infix_op } e' : t''} \quad \text{EET_APPLYI} \\
\frac{E \vdash e_1 : \mathbf{bool} \quad E \vdash e_2 : t \quad E \vdash e_3 : t}{E \vdash \mathbf{if } e_1 \mathbf{ then } e_2 \mathbf{ else } e_3 : t} \quad \text{EET_IFTHENELSE} \\
\frac{E \vdash e_1 : \mathbf{unit} \quad E \vdash e_2 : t}{E \vdash e_1; e_2 : t} \quad \text{EET_SEQUENCE} \\
\frac{E, x_1 : t_1 \vdash e : t}{E \vdash \mathbf{function } x_1 \rightarrow e : t_1 \rightarrow t} \quad \text{EET_LAMBDA} \\
\frac{E \vdash e : t \quad E, x : t \vdash e' : t'}{E \vdash \mathbf{let } x = e \mathbf{ in } e' : t'} \quad \text{EET_LET}
\end{array}$$

$$\boxed{e \longrightarrow e'}$$

$$\begin{array}{c}
\frac{}{(\mathbf{function } x \rightarrow e) \ v \longrightarrow \{v/x\}e} \quad \text{JO_RED_APP} \\
\frac{}{\mathbf{if true then } e_1 \mathbf{ else } e_2 \longrightarrow e_1} \quad \text{JO_RED_IFTRUE} \\
\frac{}{\mathbf{if false then } e_1 \mathbf{ else } e_2 \longrightarrow e_2} \quad \text{JO_RED_IFFALSE} \\
\frac{}{(); e \longrightarrow e} \quad \text{JO_RED_SEQ}
\end{array}$$

$$\begin{array}{c}
\frac{}{\mathbf{let } x = v \mathbf{ in } e \longrightarrow \{v/x\}e} \quad \text{JO_RED_LET} \\
\\
\frac{}{v = v \longrightarrow \mathbf{true}} \quad \text{JO_RED_EQ} \\
\\
\frac{}{\mathbf{true} = \mathbf{false} \longrightarrow \mathbf{false}} \quad \text{JO_RED_EQ2} \\
\\
\frac{}{\mathbf{false} = \mathbf{true} \longrightarrow \mathbf{false}} \quad \text{JO_RED_EQ3} \\
\\
\frac{e \longrightarrow e'}{C.e \longrightarrow C.e'} \quad \text{JO_RED_CONTEXT}
\end{array}$$

Definition rules: 33 good 0 bad
 Definition rule clauses: 65 good 0 bad